

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Mars has low mass so weak gravity and could not retain a dense atmosphere (1) • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures (1) • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen (1)	3
01.2	<i>1 mark for each point.</i> • Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil (1) • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured (1) • Too close (Venus) - too hot; too far (Mars) - too cold (1)	3
01.3	<i>1 mark for each point.</i> • The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface (1) • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) (1)	2
01.4	<i>1 mark for each point.</i> • The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun (1) • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
02.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ (1) • $N(N - 1) = 100 \times 99 = 9900$ (1) • $D = 9900 / 2900 = 3.41$ (accept 3.4) (1)	3
02.2	<i>1 mark for each point.</i> • Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B (1) • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) (2) • High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow (2) • Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) (2) • Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes (2)	8
03.2	<i>1 mark for each point.</i> • Predators are killed under licence to protect stock, reducing wild populations of protected / valued species (1) • Better cage designs, netting, fencing, bird scarers exclude predators without killing (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	
04.1	<i>1 mark for each point.</i> • In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis (1) • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) (1) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed (1) • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate (1)	4

Question	Marking guidance	Marks
04.2	<i>1 mark for each point.</i> • Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced (1) • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor (1)	2
04.3	<i>1 mark for each point.</i> • Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone (1) • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments (1) • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy) (1)	3
04.4	<i>1 mark.</i> • Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	
05.1	<i>1 mark for each point.</i> • Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm²) (1) • They use the polyp's CO₂ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently (1) • They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients (1) • Polyps also trap plankton with cilia and stinging nematocysts; surplus CO₂ is used to build the calcium carbonate skeleton (1)	4
05.2	<i>1 mark for each point.</i> • High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water (1) • Found only where all these coincide (about 1% of ocean area) (1) • Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance (1)	3
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Fisheries: support over a quarter of marine species including fish, lobsters, conch (1) • Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores (1) • Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts (1) • Climate control: carbonate skeletons sequester CO₂ (1) • Tourism (diving, snorkelling) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	
06.1	<i>1 mark for each point.</i> • Negative (1) • Positive (1) • Negative (1) • Positive (1)	4

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> - The response to a change acts to re-establish / return the system to the previous equilibrium (1) - So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback (1)	2
06.3	<i>1 mark for each point.</i> - eg the possible release of methane hydrate at the end of the Permian era (1) - Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium (1)	2
06.4	<i>1 mark for each point.</i> - Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) (1) - Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
07.1	<i>1 mark for each point.</i> - Sigmoidal (S-shaped) growth curve (1) - Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate (1) - Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate (1) - Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate (1)	4
07.2	<i>1 mark for each point.</i> - Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation (1) - Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter (1)	2
07.3	<i>1 mark for each point.</i> - Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density (1) - At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland) (1)	2
07.4	<i>1 mark for each point.</i> - Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population (1) - Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
08.1	<i>1 mark for the correct answer.</i> $1000 / 100\,000 \times 100 = 1\%$ (1)	1

Question	Marking guidance	Marks
08.2	<i>1 mark for each point.</i> • Some light is reflected by the leaves (1) • Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed (1) • Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) (1)	3
08.3	<i>1 mark for each point.</i> • Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) (1) • After four levels too little energy remains to support a further population (1)	2
08.4	<i>1 mark for each point.</i> • Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) (1) • Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat (1)	2
08.5	<i>1 mark for each point.</i> • Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) (1) • So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
09.1	<i>1 mark for each point.</i> • Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) (1) • Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) (1) • Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions (1) • Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) (1)	4
09.2	<i>1 mark for each point.</i> • Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) (1) • Islands provide roosting / nesting sites safe from terrestrial predators (foxes) (1) • Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks (1)	3
09.3	<i>1 mark for each point.</i> • Mobile species (birds, flying insects) colonise easily (1) • Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially (1)	2

Question	Marking guidance	Marks
09.4	1 mark. Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions - Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound - Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation - Physiological research using wildlife species; wildlife as pest control agents (ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, Bt) - Genetic resources: crop wild relatives, Vavilov centres, gene pools - Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision - Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) - Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
	Model answer - Level 3 (8 marks)	
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, <i>Bacillus thuringiensis</i> - and crop wild relatives in Vavilov centres hold the genes future crops will need.	

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	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
10.2	<i>1 mark.</i> • The variety and abundance of life (species, genes, habitats) in an area / on Earth (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity • Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems • Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	25